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Dynamical representation of odors by oscillating and evolving neural assemblies

Gilles Laurent

Although smells are some of the most evocative and emotionally charged sensory inputs known to us, we still understand relatively little about olfactory processing and odor representation in the brain. This review summarizes physiological results obtained from an insect olfactory system and presents a functional scheme for odor coding that is compatible with data from other animals, including mammals. This coding scheme consists of three main and concurrent odor-induced phenomena: 20–30 Hz oscillatory mass activity; patterned and odor-specific neuronal responses; and transient, dynamic synchronization of odor-specific neural assemblies. When these phenomena are considered together, odors appear to be represented combinatorially by dynamical neural assemblies, defined partly by the transient but stimulus-specific synchronization of their neuronal components.

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A REMARKABLE feature of olfactory systems is that their anatomical design (circuit macro- and microarchitecture, dendro–dendritic synaptic arrangements and neuronal morphologies) is exceptionally similar across animal phyla – from mollusks to insects, crustaceans and vertebrates[1–4]. Therefore, the functional principles that can be established from studying one system might apply to the others as well, because similar circuit design principles are likely (although by no means certain) to underlie similar functional or computational principles. In all cases but that of pheromonal olfaction, the problem to be solved by the brain is the specific representation of 'unpredictable' stimuli; that is, ones that, in the natural world, are complex and infinitely varied multicomponent blends[5]. Because odors are so varied, olfactory systems must be able to learn the specific patterns that make each smell what it is (for example, a rose or a

Gilles Laurent is at the California Institute of Technology, Biology Division, 139-74, Pasadena, CA 91125, USA.

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Box 1. The locust olfactory circuits

Each antenna contains about 50 000 cholinergic and excitatory olfactory receptor neurons (ORNs)[a]. Each ORN sends an axon to the ipsilateral antennal lobe (al) (Fig. A,B), where it terminates in one, two or three of the 1000 glomeruli. A glomerulus thus contains the axonal terminals of 50–150 ORNs. There is good evidence in the pheromonal system of moths that receptor neurons of the same molecular sensitivity converge on to the same glomerulus[b]. However, it is not known (in any insect) whether the non-pheromonal afferents that converge into any one glomerulus express the same olfactory receptor gene(s) (as yet unidentified), as has been demonstrated recently in the vertebrate olfactory bulb[c–e].

The antennal lobe contains two main types of neuron: the local (LNs) and projection neurons (PNs)[a,f,g]. The local neurons (about 300 in each lobe[a]) are axonless and send dense projections throughout the entire antennal lobe neuropil. Locust LNs do not generate conventional Na^+ action potentials but rather TTX-resistant (probably Ca^{2+}-dependent) spikelets of variable shapes and amplitudes, which can be recorded from the soma or the dendrites[g]. Spikelets of different shapes and amplitude can often ride on one another, indicating that they can be generated independently at multiple sites within each dendritic tree[g]. LNs are GABAergic and inhibitory. Immunogold histochemistry indicates that GABAergic profiles (presumably belonging to LNs) make output synapses on to both GABAergic and non-GABAergic profiles, and that they receive synaptic inputs from both GABAergic and non-GABAergic profiles[h]. Therefore, LNs probably receive direct synaptic input from ORNs, PNs and from other LNs, and synapse on to PNs, LNs and possibly also on ORN terminals, as supported by data from other insects[i,j].

The projection neurons (about 830 in each lobe[h]) each arborize in 10–20 glomeruli[k], and thus resemble the mitral cells of lower vertebrates rather than of mammals. Each PN sends an axon in a common tract towards the base of the ipsilateral mushroom body calyx (Fig. C) and ends in the lateral protocerebral lobe[f] (LPL). Each PN axon sends several collaterals in passing that diverge widely within the mushroom body calyx (as mitral/tufted axons do in piriform cortex), and thereby preserve no precise spatial register of their origin. These branches form large varicose terminals that contact the spiny dendrites of the intrinsic interneurons of the mushroom body, the Kenyon cells (KCs). PNs are cholinergic, excitatory and conventionally spiking neurons. They make output synapses both in the antennal lobe (on to other PNs and LNs[h]), and in the calyx of the mushroom body (as well as, presumably, in the LPL).

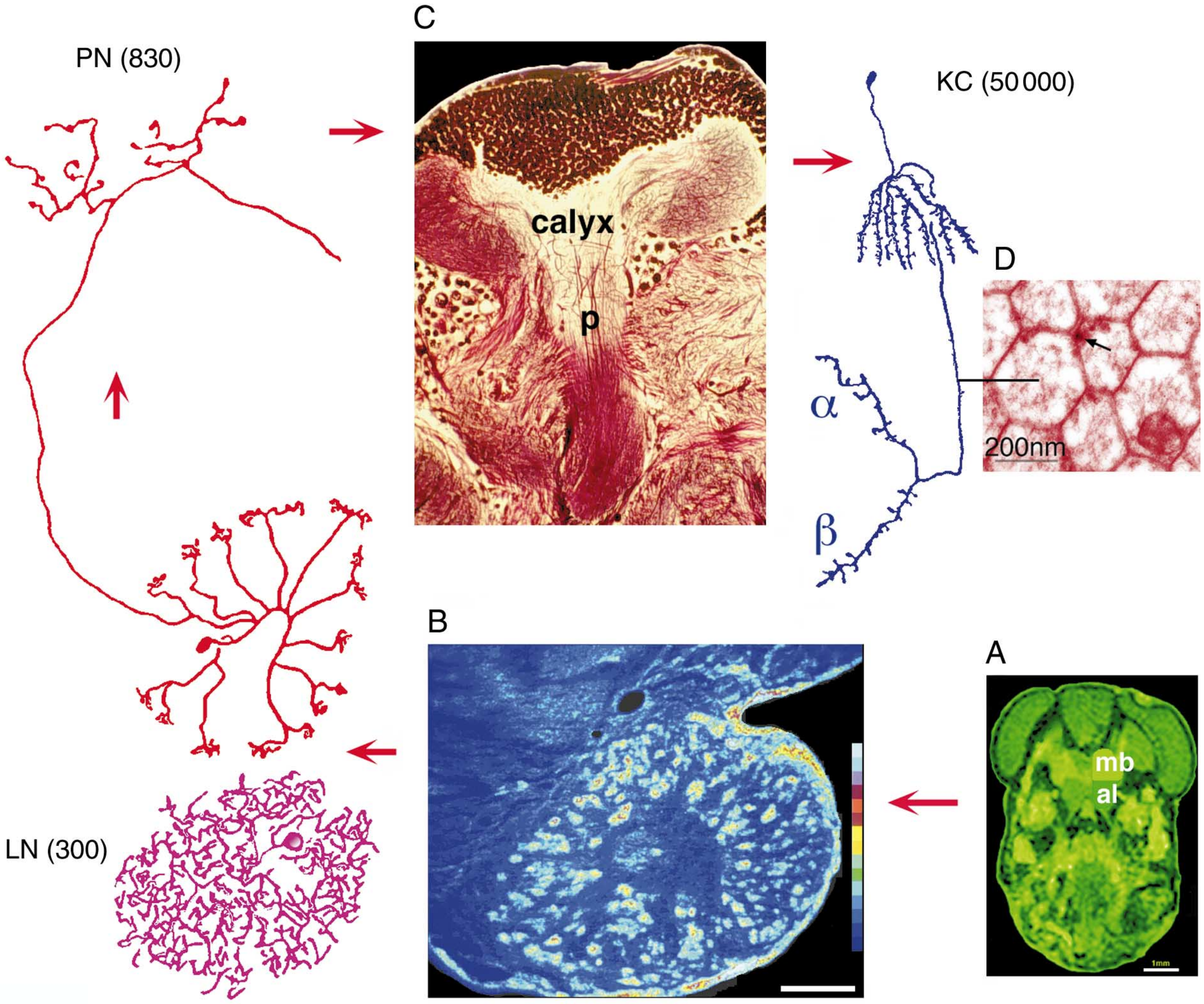
Each KC (of which there are about 50 000 in each mushroom body) forms a field of spiny dendrites (pure input sites[h]) within the mushroom-body calyx. Physiological evidence indicates that each KC can receive convergent inputs from at least 10–20 PNs (Ref. k; G. Laurent, unpublished observations). All KCs send an axon ventrally, collectively forming the pedunculus (P) of the mushroom body. All KC axons are closely juxtaposed, form tight hexagonal arrays devoid of glial wrapping[h] (Fig. D) and bifurcate into the α and β lobes, where they terminate. Kenyon cells form output synapses (often reciprocally), both in the pedunculus[h] (with adjacent KC axons; see Fig. D) and in the lobes (with adjacent KCs and with extrinsic neurons). Kenyon cells are not GABAergic and their neurotransmitter(s) is (are) as yet unknown. Their action is believed, although not clearly demonstrated, to be excitatory. Their dendrites express voltage-gated inward currents[k], which can greatly amplify the excitatory synaptic inputs that they receive from PNs (see Fig. 1C,D and text).

Fig. ***Summary of the locust olfactory pathways.*** **(A)** *Magnetic resonance image of a locust head (12T magnet) (obtained in collaboration with H. Davidowitz and R. Jacobs). The positions of the antennal lobe (al) and mushroom body (mb) are highlighted on one half of the brain.* **(B)** *Fluorescence image of the antennal lobe showing the glomeruli [intense (yellow–red) clusters, each approximately 10–40 μm in diameter] in one plane of section. The local neurons (LNs) and projection neurons (PNs) were stained by intracellular injection of cobalt and drawn with a camera lucida from whole mounts. They are represented at the same scale as the micrograph in* B *to show the extent of their arbors.* **(C)** *Photomicrograph of the mushroom body (same scale as* B*). This is a Bodian sliver-stain specimen kindly provided by H-J. Pflüger. The pedunculus (p), or tract containing the axons of the Kenyon cells (KCs), is marked on the micrograph. Note the densely packed KC somata dorsally. All the dendrites of the KCs have spines and collect inputs from the PNs in the calyx. A KC has also been drawn from a whole mount of an intracellular cobalt stain (same scale as* B, C *and the drawings of the PN and LN).* **(D)** *High-power electron micrograph of KC axons sectioned transversally in the pedunculus[h]. Note the tight hexagonal array and the synapse (arrow) formed by one axon on its neighbor. (Work in collaboration with B. Leitch.) Scale bars in* A, B *and* C*, 100 μm.*

fertilizer). They must form unique representations, store them (often for the entire lifetime of the animal) and allow identification (recall) even when the odor is slightly different from that learnt originally (pattern completion)[6]. This review focuses on recent work on olfactory processing by locusts and presents a hypothesis for odor coding by the projection neurons (PNs) of the antennal lobe (the functional analog of mitral/tufted cells in the vertebrate olfactory bulb) that might apply to other olfactory systems as well.

Macroscopic (oscillatory) events generated by odors

When a monomolecular odorant (for example, an aliphatic ketone) or a blend (for example, a fruit or leaf fragrance) is air-puffed on to one antenna of a locust, extracellular potentials recorded in the antennal lobe (Box 1) reveal a long-lasting, negative, local field potential (LFP) that is indicative of a barrage of excitatory afferent input[7]. Extracellular recordings from the mushroom body calyx, however, reveal oscillatory LFPs with a frequency of 20–30 Hz (Fig. 1A,B). These oscillations indicate the presence of an odor but the signal frequency is independent of, and therefore contains no information about, the nature of the odor. The LFP oscillations last for the duration of the odor puff or slightly longer, and occur without a phase gradient over the entire mushroom body calyx[8]. Intracellular recordings from Kenyon cells (the intrinsic neurons of the mushroom body, Box 1) during an odor response reveal regular, depolarizing, synaptic events phase-locked to the LFPs (Ref. 7; Fig. 1C). These synaptic events are EPSPs, the amplitude of which can be artificially amplified by DC intracellular depolarization, indicating voltage-gated inward currents in the Kenyon cell dendrites. These amplifying events can also be triggered by the summation of EPSPs elicited by electrical stimulation of presynaptic PNs in the antennal lobes (Fig. 1D). Therefore, a Kenyon cell generally responds to a given odor by producing action potentials whose timing is locked to the LFP



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oscillation (Fig. 1C). Although most sampled Kenyon cells display subthreshold, rhythmic, synaptic patterns during an odor, very few produce action potentials like those recorded in Fig. 1C. Therefore, many converging inputs from many PNs seem to be required to elicit a true (suprathreshold) Kenyon cell response, and Kenyon cells responding to any one odor are probably relatively sparse.

Where do the LFP oscillations originate? If a spectral analysis is carried out on a mushroom body LFP in the absence of an odor, a small but detectable amount of power is seen in the 20–30 Hz band. Therefore, it is possible that the mushroom body circuits are 'tuned' intrinsically to oscillate and that oscillations are the result of mushroom-body circuit dynamics. Three results, however, indicate that the oscillations are driven primarily by the antennal lobe.

(1) If oscillations were caused only by mushroom-body circuit dynamics, then visual inputs, which also converge on to the mushroom body, would also be expected to cause coherent oscillations there; this is not the case. If stepped visual stimuli are presented to the animal, mushroom-body LFPs show clear DC shifts precisely correlated with the stimuli but no change in power in the 20–30 Hz range.

(2) If the antennae are cut or if the antennal lobes are severed, mushroom body LFPs lose all detectable power in the 20–30 Hz range.

(3) As detailed below, oscillations occur in antennal lobe neurons, even after ablation of the mushroom body. This indicates that the oscillations observed in the mushroom body LFPs are driven by the coherent activity of PNs carrying odor information from the antennal lobe. However, this result does not eliminate the possibility that the mushroom body circuits are tuned independently to oscillate at the same frequency and, therefore, to 'accept' the synchronized and rhythmic input from the antennal lobe.

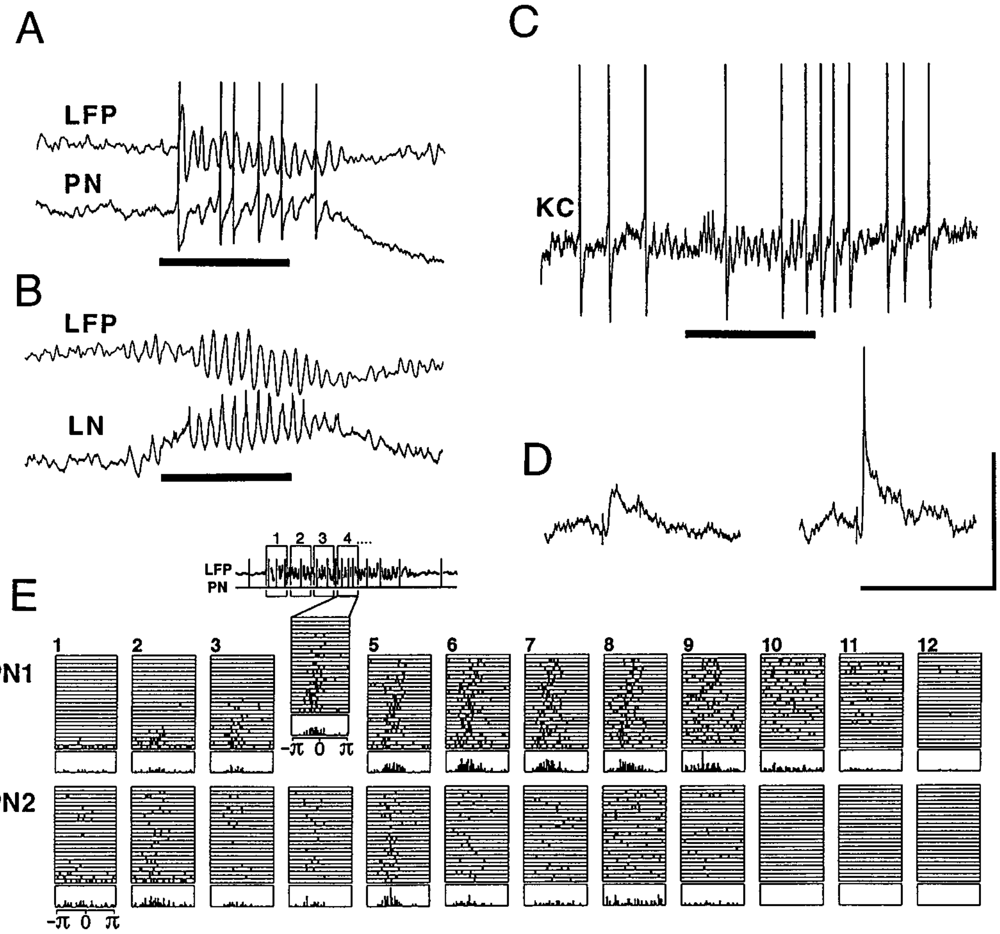

Fig. 1. ***Oscillatory responses and transient synchronization of antennal lobe and mushroom body neurons during odor responses in intact locusts.*** **(A)** *Simultaneous local field potential (LFP) recording from the mushroom body and intracellular recording from an antennal lobe projection neuron (PN). Note the odor-elicited 20 Hz LFP oscillations, the synchronization of the PN to the LFP, and the PN subthreshold oscillations composed of alternating EPSPs and IPSPs. Bar: 0.5 s pine odor puff; Scale bar: horizontal, 0.5 s; vertical, 0.4 mV (LFP) and 10 mV (PN).* **(B)** *Simultaneous LFP recording from the mushroom body and intracellular recording from a dendrite of an antennal lobe local neuron (LN). Note the LN membrane potential oscillations and their synchronization to the LFP. Bar: 0.5 s apple odor puff; Scale bar horizontal, 0.5 s; vertical, 0.4 mV (LFP) and 10 mV (LN).* **(C)** *Intracellular recording from the soma of a Kenyon cell (KC) in the mushroom body. Note the odor-elicited rhythmic EPSPs and the action potentials that some of them lead to. Bar: 1 s strawberry odor puff; Scale bar: horizontal, 1 s; vertical, 10 mV.* **(D)** *EPSPs caused in KCs by electrical stimulation of the antennal nerve. These EPSPs are mediated by direct inputs from the antennal lobe PNs (Ref. 7), and can be amplified by intrinsic voltage-gated currents in the KC dendrites. (Left) Low amplitude stimulus; (Right) higher amplitude stimulus, which causes summation and active boosting of the EPSP. The sharp potential riding on the EPSP is not an action potential but a graded active response. Scale bar: horizontal, 125 ms; vertical, 10 mV.* **(E)** *Phase diagrams showing the transient synchronization of PNs to the LFP. The response of each of two PNs (PN1 and PN2) to an odor has been divided into 12 successive epochs (1–12). In each epoch, the phase of every action potential relative to the corresponding LFP cycle was measured, as indicated for epoch 4. The phase was plotted from $-\pi$ to $+\pi$, where 0 corresponds to the positive peak of the LFP oscillation. This operation was repeated for 24 odor trials (rasters), and the numbers of spikes counted in each phase bin are plotted in histograms under each epoch. It is obvious that PN1 was very clearly synchronized to the LFP during epochs 3–8 but not during epochs 10–11. Similarly, PN2 was tightly synchronized to the LFP during epoch 5 but not at all during epoch 8, although a similar number of spikes was produced during both epochs.*

Odor-specific neural assemblies

Odors, whether simple or blended, appear to be represented combinatorially by overlapping ensembles of PNs. Although the responses of PNs are complex (see details below), many PNs can always be found, in the same animal, to show a reliable modification of their background activity upon presentation of a chosen odor[9]. A given PN often responds to several odors[8]; hence, the representations of different odors overlap. In the course of an experiment, about 10% of the neurons sampled generally 'respond' to any odor tested (over 2 log units of odor concentration). This suggests that about 100 PNs (out of 830) are activated by the presentation of an odor. By recording the activity of PNs intracellularly, it is possible to determine not only their response but also to infer the responses of other antennal lobe neurons that drive them, thanks to a generally clear, odor-elicited, synaptic background (Fig. 1A). This is most useful when considering odor blends[8]. From such experiments, it appears that knowing the response (sub- and suprathreshold) of a PN to two odors does not enable its response to the 1:1 blend of these two odors to be predicted[8]. Therefore, it seems that the response of PNs to odors results not only from the input originating in activated olfactory receptor neurons but also from complex local interactions and circuit dynamics within the antennal lobe. Another possible (and non-exclusive) explanation lies in the complex integrative properties of the olfactory receptor neurons themselves, which, in some animals, can be excited by certain odors and inhibited by others[10]. Therefore, the drive that a given receptor neuron produces to PNs and local neurons (LNs) is likely to depend not only on the presence of the optimal molecular stimulus but also on the molecular 'background' in which this stimulus occurs (that is, the odor blend).

Slow temporal response patterns: dynamical assemblies

Although some PNs can nearly always be found to produce a reliable stimulus-induced change in their firing behavior, responsive PNs do not all respond in a similar way to the same stimulus. If the duration of the 'ensemble response' is defined as the duration of the LFP oscillations in the mushroom body, some PNs produce action potentials mainly at the beginning of the ensemble response, others at the end, and yet others during several epochs (for example, beginning and end, but not middle) of the ensemble response[9] (Fig. 1E). Prominent periods of (apparently synaptic) inhibition often occur during the periods of silence surrounding, preceding or following activity. Although these patterns vary across neurons, they are consistent for each neuron and odor (at the same concentration). In other words, 30 or more successive trials with odor A produce the same qualitative response in the same PN (although never the same exact number of action potentials), whether trials with odor A are intercalated with ones with another odor or not[9] (Fig. 1E). These patterns of responses are odor-specific as well as neuron-specific. Therefore, a PN can (but does not necessarily) respond to two or three different odors of those tested but displays different firing patterns in response to each odor[8,9]. Odor-specific 'ensemble responses' can thus be defined as spatially distributed representations that evolve in time (dynamical patterns) over the duration of the response. In other words, the odor-specific ensemble that represents an odor changes in a reliable fashion and consists, in fact, of a precise succession of neuronal assemblies.

Fast temporal features: transient synchronization

Odors elicit oscillatory population responses in the mushroom body and it was suggested above that the oscillations are driven by activity in the ipsilateral antennal lobe (that is, by the PNs). This can be demonstrated directly from intracellular recordings of PNs or LNs (Fig. 1A,B). The clearest 'unprocessed' evidence comes from LNs, which show clear membrane potential oscillations (some of which give rise to active but TTX-insensitive potentials or 'spikelets') that are precisely phase-locked to the mushroom body LFP (Fig. 1B). Like PNs, individual LNs only respond to certain odors; therefore, these oscillatory patterns are odor-specific in each LN. Projection neurons also produce oscillatory responses but the demonstration of these requires careful analysis.

(1) When spike production is prevented by hyperpolarization, the PN membrane potential shows clear alternating EPSPs and IPSPs whose timing is phase-locked to the LFP during odor responses[8] (Fig. 1A).

(2) The timing of these recurrent IPSPs relative to the LFP oscillations corresponds precisely to that of the peak depolarization of the LNs (Ref. 8).

(3) The timing of the PN depolarizing peaks [and thus of the spike(s) each produces] lags the IPSPs by about 90°.

(4) Ablation of the mushroom body calyx does not alter these oscillatory patterns in the antennal lobe neurons. [These second, third and fourth points are consistent with the theory that the oscillations result primarily from odor-induced network interactions between groups of LNs and PNs (see Box 1), rather than from intrinsic properties of these neurons or from feedback from the mushroom body.]

(5) The phase of LN or PN activity relative to the LFP is independent of the odor. Thus, information about the stimulus is not contained in the phase of PN spikes, as measured relative to the LFP. This suggests that the odor-encoding neuronal ensembles might be defined by their synchrony of firing and, thus, that synchronous oscillations of pairs of PNs or LNs should be observable and odor-specific. This hypothesis was demonstrated directly by making paired intracellular recordings of LNs and PNs and calculating the cross-correlation between their membrane potentials over single stimulus presentations[8,9] (Fig. 2).

The stimulus representation contains yet another level of complexity. The fact that a PN produces action potentials during a part of the ensemble response does not mean that all of these action potentials are phase-locked to the field potential. Rather, it often happens that the action potentials produced by a PN during a relatively precise epoch (for example, the window between 400 and 650 ms after the stimulus onset) consistently fail to phase-lock to the LFP (Refs 8,9). This can be shown by measuring the phase of PN action potentials stage by stage over sufficiently small epochs that make up the total response (Fig. 1E). Only then does it become apparent that although PNs do tend to oscillate and phase-lock to the ensemble LFP, they often do so only during certain epochs. For example, PN2 in Fig. 1E produced, on average, as many spikes in epoch 8 as it did in epoch 5 but the spikes in epoch 8 were not phase-locked to the LFP whereas those in epoch 5 were. The epochs during which non-phase-locked spikes occur in one PN can be found at any time during the ensemble response but usually happen at a time of transition between activity and silence (or vice versa) for this neuron. These patterns of locking and unlocking are also odor-specific. Therefore, cross-correlation analysis must be carried out stage by stage (for example, using a sliding window[8,9]; Fig. 2) to reveal any transient bouts of synchronization between pairs of neurons that would often go undetected by conventional, time-averaged, cross-correlation analysis[9].

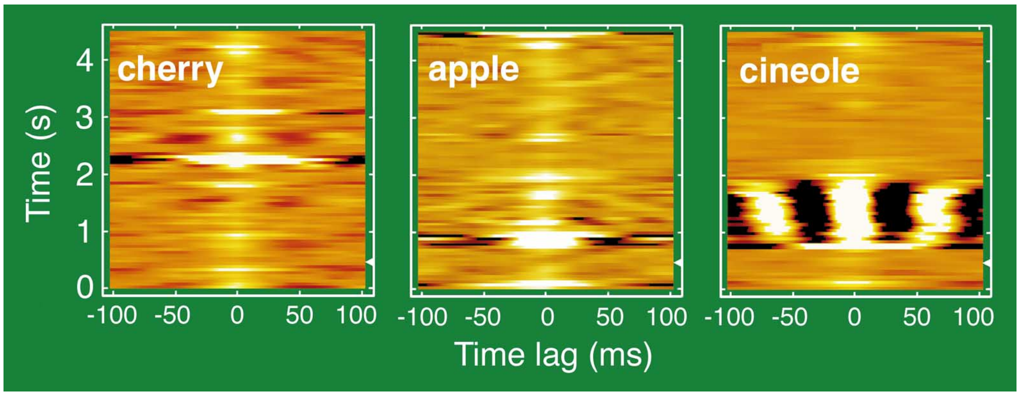

Fig. 2. ***Odor specificity of neuronal synchronization.*** *The cross-correlation function between the membrane potentials of two local neurons (LNs) of the antennal lobe was calculated using a sliding window[8] during a single presentation of each of three odors (cherry, apple and cineole). The cross-correlation obtained for each window was color-coded and all correlation functions were stacked on one another. As a result, each diagram represents the evolution of the cross-correlation between the two LNs. Odors were presented between –500 and 500 ms. The delay in response was due to the physical delay in the odor delivery. The two LNs tested oscillated and synchronized to one another in response to cineole only.*

A functional odor-coding scheme

When an odor is presented to the animal, a group of PNs [perhaps about 100 out of 830 (Refs 8,11)] is activated (Fig. 3). Not all activated PNs respond in the same way or at the same time, so that the representation is distributed (or evolves) in time. So, during any one epoch a subset of PNs is active; many of these PNs fire coherently (their cross-correlation function is periodic and its central peak shows no major phase-lag) at a frequency of about 20–30 Hz. It is this subset of PNs that, by its direct action on the dendrites of Kenyon cells in the mushroom body, causes the current loops that can be detected as LFP oscillations (Fig. 3). For example, in the first highlighted epoch in Fig. 3, the odor is represented and the LFP oscillation is caused by PNs 1,2,4,7,9 and 11. These PNs diverge widely in the mushroom body and contact many

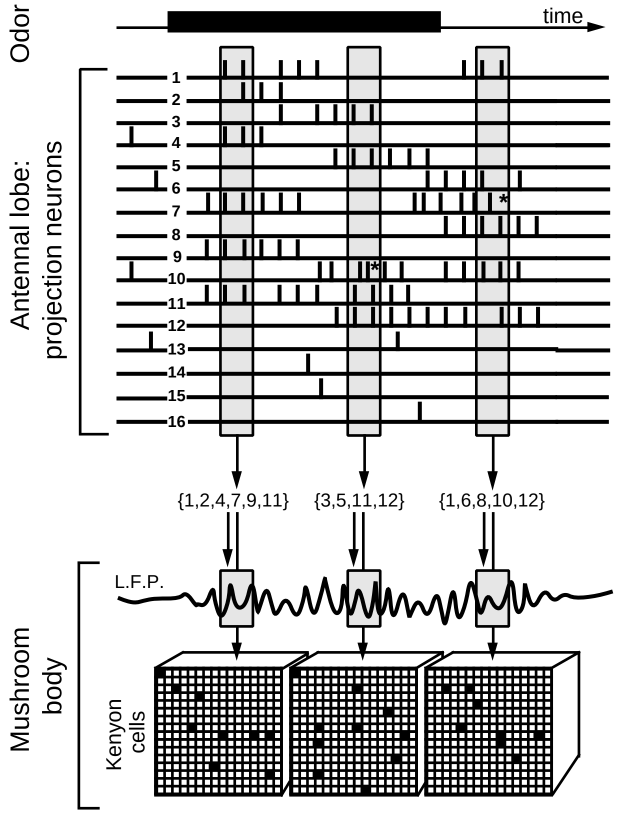

Fig. 3. ***The odor-elicited neural events in the locust olfactory system.*** *The presumed odor representation is combinatorial, spatially distributed and relies on synchronized and evolving neural assemblies. See text for details. Asterisks indicate non-phase-locked action potentials.*

Kenyon cells. However, only a few of these Kenyon cells will receive enough converging PN inputs to amplify the spatially summed EPSPs via their active, depolarizing, dendritic conductances and trigger an action potential. These few activated Kenyon cells are probably the ones that associate the PN input patterns and that ultimately come to encode the odor signal at this time (Fig. 3).

Some time later (for example, the second highlighted window in Fig. 3), the subset of active PNs has changed although it might overlap with the previous one. In this proposed scheme, only the PNs that are synchronized (that is PNs 3,5,11 and 12, but not 10) encode the stimulus. It is this new assembly that now causes the LFP oscillation (of the same frequency) and activates a new (and possibly overlapping) set of Kenyon cells. In other words, the continuous LFP oscillations fail to reveal that the ensemble of neurons that generates them continuously evolves. But why should only the synchronized PNs be counted and not those that do not phase-lock in each epoch? After all, non-synchronized PNs also contribute to the response of Kenyon cells by depolarizing them. The main reason for this is that Kenyon cell dendrites have prominent, voltage-gated, depolarizing conductances; that is, a biophysical device that specifically selects and amplifies synchronized inputs. In other words, a PN has a great 'advantage' in being synchronized with other PNs because only then will its effect have a high probability of being 'felt' by the postsynaptic Kenyon cell. But then why should a PN spike at all if its action potentials, at one particular time, cannot be synchronized to the population? The reason might be that it cannot be helped. Indeed, the network phenomena that underlie the dynamical response of each PN probably involve fast and slow synaptic and intrinsic phenomena, so that, for example, the transitions between periods of activity and lack of activity might sometimes require progressive, rather than sudden, synchronization or desynchronization. Therefore, it is possible that the serial formation of odor-specific ensembles, first in the antennal lobe and then in the mushroom body, is a strategy to refine the odor representation and to make it more sparse. As a result of such a mechanism, the memory capacity of the mushroom body would be increased by reducing the overlap between representations.

Could these principles be more general?

Are these three odor-induced phenomena (20–30 Hz oscillatory mass activity, patterned and odor-specific neuronal responses, and transient, dynamic synchronization of odor-specific neural assemblies) observed in other olfactory systems? If so, are they concurrent, thus defining an apparent dynamic code for odors in these systems?

'Induced waves', or EEG oscillations caused by olfactory stimulation, have been observed in the olfactory bulb of amphibians[12], fish[13,14] and mammals[15–17], including humans[18]. These oscillations are usually in the γ frequency range (30–60 Hz), and are often modulated by slower oscillations (Θ range, 5–12 Hz) coupled to breathing. In rats, for example, multiple simultaneous bulbar EEG recordings reveal (after much appropriate processing of phase and root-mean square amplitude) spatial patterns of oscillations that are odor-specific – but only if the odors are presented within a behaviorally significant context[19,20]; in contrast, the observations made in locusts do not reveal such context-dependent responses. Oscillations have also been observed in the olfactory system of a terrestrial mollusk (*Limax maximus*)[21–23]. The observations of odor-induced oscillations in locust reported here have also been noted in other insects, specifically cockroaches and honeybees (M. Stopfer and G. Laurent, unpublished observations). In all cases, oscillatory activity is triggered, or greatly enhanced, by odor stimulation. In *L. maximus*, LFP oscillations of similar amplitudes to those of other species occur in the absence of olfactory stimuli but odors cause a collapse of the phase gradient that exists at rest across the procerebral lobe[22,23]. In all cases, the oscillations appear to result from local interactions between inhibitory local and excitatory PNs (within the bulb of vertebrates[20], the procerebral lobe of *L. maximus*[23] or the antennal lobe of insects[8], although it has never been conclusively proven), and to reflect the coherent activity of large numbers of these neurons.

All olfactory bulb responses examined to date also provide evidence for odor-induced temporal patterns, consisting at least of sequential excitatory and 'suppressive' phases of neural activity. These patterns have been observed in amphibians[24,25], fish[26,27] and mammals[28,29]. It has been suggested that 'suppression' patterns arise in amphibian olfactory bulb neurons when odor concentrations are too high ('concentration-tuning hypothesis'[24]). However, recent data from mammals indicate that suppression patterns are not more frequent at higher odor concentration, and that certain neurons switch from an inhibitory to an excitatory response as odor concentration is raised[29]. In cockroaches[30] and moths[31], temporally structured responses can be observed in response to pheromones. Therefore, temporally patterned responses might exist in neurons in the first olfactory relay of many animal phyla and classes.

The coexistence of oscillatory mass activity and temporal activity patterns in individual neurons of the animal groups listed above suggests that the dynamic representation proposed here for the locust might also exist in many other animals. The work summarized in this article is the first in which all the micro- and macroscopic phenomena have been observed together and appear to 'interlock' in a coherent fashion, showing the transient synchronization of neural assemblies and progressive transformation of coherently active populations. The separate pieces of evidence from vertebrates and mollusks seem to be compatible with our coding hypothesis in the olfactory system but this remains to be tested.

Are oscillations useful?

It is possible that oscillations are not used for coding but are simply a by-product of the functional architecture of olfactory networks; that is, networks with prominent negative feedback loops[32]. This hypothesis is plausible, given the similarities between the microarchitectures of the primary olfactory relays of insects, mollusks and vertebrates and the probable importance of negative feedback loops for gain control[3] and tuning[33]. A mechanistic study of these oscillatory phenomena is required to determine whether the mechanisms causing oscillatory synchronization are indeed inseparable from those responsible for tuning odor responses.

A second, more interesting, possibility links synchronization to learning. One important function of this olfactory center might be to exploit synchronous activity patterns to form memories of odors (which are, by nature, combinatorial), using coincidence-dependent learning rules[34,35]. Although LFP oscillations generally result from the synchronization of large numbers of neurons, synchronization *per se* does not need to be periodic[36]. However, given the nature of physiological systems it is probably easier to generate periodic than non-periodic synchrony. Thus, oscillations would only be 'useful' in the sense that they underlie coincident activity, which is important for certain forms of plasticity. This hypothesis suggests that the formation of olfactory memories requires a particular combination of neuronal architecture, activity and synaptic physiology in a scheme similar to that hypothesized for the piriform cortex[35,37]. The prediction that plasticity in the antennal lobe or the mushroom body, or both, uses biophysical and synaptic coincidence detector mechanisms with a fine temporal resolution needs to be tested. Although, no NMDA-receptor-like channels, or channels with the same associative properties, have yet been described in insect brains, second-messenger pathways, which might also play this role, do abound[38,39].

A third hypothesis is that the existence of oscillations creates a new variable (phase) with which to encode information. Indeed, the existence of oscillations defines a periodic 'clock' signal, in relation to which the phase of individual action potentials might be measured[40]. Alternatively, phase-lags might be measured between neuronal groups, rather than between individual neurons and the average population activity (field oscillations). Many theoretical arguments favor such mechanisms for purposes as diverse as segmentation (the separation of stimulus features such as object from background) in vision or hearing[36,40,41] or concentration-invariant odor perception in olfaction[42]. Although segmentation of odors is notoriously difficult in humans, it is known that honeybees can 'perceive' the individual components of certain binary mixtures[43]. It is at present improbable, from the work described here or from other data, that reliable and stimulus-specific phase-lags exist between antennal lobe neurons in the olfactory system, although this hypothesis should continue to be tested directly using multiunit recordings.

A fourth hypothesis is that oscillations might permit the refinement of stimulus representation in two sequential stages, by filtering out 'unwanted' action potentials between the first and second relay. In the example described in this article, the brain would first build large distributed odor representations in the antennal lobe, using a large subset of the available neurons (about 10%), and then form sparser representations, less prone to overlap and generalization, in the mushroom body, where storage is probably implemented[38,44,45]. This selective filter would exploit the dendritic, voltage-gated non-linearities of Kenyon cell dendrites, as explained earlier. This hypothesis is reminiscent of, but different from, Walter Freeman's idea whereby, because of its own internal dynamics, the mammalian piriform cortex identifies as the signal from the olfactory bulb only the input that is coherent, and defines everything else as 'noise'[46]. Problems caused by temporal dispersion of the inputs, which are significant in the mammalian olfactory system, are much less likely to be significant in insect brains, which are very small.

A fifth hypothesis relates to the problem posed by distributed, combinatorial neuronal representations. If a stimulus (such as an odor) is represented by an ensemble of neurons (rather than by one highly specific neuron), this representation can be formalized as a vector sum of all the vectors contributed by each participating neuron[47]. In this representation, the space where stimuli lie has as many dimensions as there are component neurons (about 830 here, assuming independence between PNs), and each odor can be represented by either a point or a small cloud (if the ensemble is static) or a trajectory (if the ensemble is dynamic) within that space. The contribution of each neuron (that is, the length of the vector contributed by each neuron) is measured by the number of spikes it provides to the ensemble. Of course, this length depends on the time over which spikes are counted. If the stimulus or its representation is (as described in this article) a dynamical one, a clock signal is needed to update this population vector as both the stimulus and the response to it unfold. The oscillations might provide this clock signal with an added advantage. When a neuron synchronizes to the population, it usually produces only one action potential per oscillation cycle. In other words, the contribution of each vector to the population vector is 0 (no spike or a non-synchronized spike) or 1 (one synchronized spike) at each oscillation cycle. This has the possible advantage of constraining the population vector trajectory to the edges of an 830-dimensional cube, rather than allowing it to visit an infinitely large space. Therefore, in this scheme, the oscillations would define the clock according to which vector direction is updated, and simultaneously limit the possible length of this vector and of its trajectory. This type of reduction of the 'coding space' might be very important in the design of an optimal memory system whose role is to store new, and recognize old, distributed odor representations.

Why dynamical representations?

Because the dynamical features described here are odor-specific, I propose that they participate in the encoding of the stimulus. What advantage would a dynamical representation offer over a static one? The nature of odors is such that their molecular components, as opposed to, for example, the spatial features of a face or the temporal features of speech or a melody, cannot be separated or discriminated in space or time. This results indirectly in our generally poor ability to describe odors precisely and accurately. Thus, an odor is a specific, multidimensional, but 'amorphous', pattern that must be learnt and later recognized. It is possible that imposing a temporal structure on the odor representation offers a way to present simpler (that is, containing less simultaneous components) successive representations or patterns to the networks that are in charge of learning and later recognizing them. Although each successive neural assembly in the dynamical representation (Fig. 3) probably has nothing to do with the representation that each component of a blend would elicit on its own, it is nevertheless a way to impose, via neural means, an artificial but reliable temporal 'dissection' of the stimulus. The task of the network might then be to learn each 'image' in this evolving representation, or even possibly the precise sequence (or portions of this sequence) in which these 'neuronal images' follow each other. It is known that speech or song-specific systems learn and recognize specific sequences of sounds using neurons that are exquisitely sensitive to the order of syllables[48]. Therefore, it is possible that the olfactory system imposes a temporal encoding of the olfactory stimulus so that it can use sequence detectors for pattern learning and recognition.

Concluding remarks

Many of the hypotheses presented here are compatible with each other. The remaining experimental problems that must be solved to test these hypotheses, however, are formidable, and must involve a combination of refined physiological, behavioral and psychophysical techniques. For example, it would be most useful to be able to ask an animal: 'Did you smell and recognize odor A?' under conditions in which firing synchrony or the slow dynamical patterns, but not the identity of the active neurons, are selectively disrupted. The tools for such experiments are, unfortunately, still undefined; it is the construction of such tools that, to my mind, is one of the most challenging tasks for present and future neuroethologists and brain scientists interested in coding the brain. We now know that reliable and stimulus-specific temporal patterns of neuronal activity occur in the brain, strongly indicating that temporal codes (whether they occur on top of a coherent oscillation or not) have a role to play in brain function. How these codes might be read, however, still remains to be determined.

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